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CASE REPORT

Sudden death due to malignant hyperthermia with a mutation of RYR1: autopsy, morphology and genetic analysis

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Abstract Malignant hyperthermia (MH) is an autosomal dominant disorder characterized by abnormal calcium homeostasis in skeletal muscles in response to triggering agents. Autopsy, morphology, and genetic analysis were performed on a 19-year-old man who died rapidly after exposure to sevoflurane during maxillofacial surgery. Muscle spasm around the operation area and limb rigidity occurred and renal tubules full of myoglobin casts were observed by microscopy. Ultrastructural changes in the skeletal muscles and the myocardium were detected by transmission electron microscopy (TEM). Genetic analysis disclosed a ryanodine receptor type 1 (RYR1) gene mutation and a nucleoide mutation in chromosome 19q (G1021A) in the deceased and his father. According to the fore mentioned results, the relationship between the cause of death and MH was confirmed. Thus, genetic analysis can be an important procedure in diagnosing MH.

Keywords Forensic pathology · Malignant hyperthermia · Molecular autopsy · Mutation of RYR1 · TEM

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Introduction

Malignant hyperthermia (MH) is a pharmacogenetic disorder of the skeletal muscles characterized by muscle contractions and life-threatening hypermetabolic crisis, which is often triggered by exposure to certain general anesthetics and depolarizing muscle relaxants. With extensive use of inhaled anesthetics and depolarizing relaxants, the morbidity and mortality rate of MH and death have visibly increased from 1:20,000 to 1:10,000 after it was first reported in 1960 [1, 2]. It is considered an autosomal dominant myopathy associated with abnormalities of the ryanodine receptor type 1 gene (RYR1) [2, 3]. The reported cases were based on a drastically elevated body temperature, myoglobin-positive material in the renal tubules, increased creatinine kinase, and genetic analysis. Postmortem genetic analysis has only been performed on a few cases [4, 5], and a mutation of the proband and his relatives has not been previously reported.

Herein, we report on a 19-year-old man who developed MH and died rapidly. Relevant examinations of the deceased revealed nucleotide changes and amino acid substitutions, and the same mutations were detected in his father's DNA after genetic analysis of both parents and his brother. Consistent with the current state of knowledge, the relationship between death and MH is associated with genetic changes and determined by genetic analysis.

Case report

Clinical course

A 19-year-old male was admitted to hospital suffering from pain and swelling of the maxillofacial region after a traffic accident. A 3 cm long laceration and fracture were present at the middle of the palate. The neck and shoulders were swollen, leading to severe limitations in motion. Skull and mandible fractures were revealed by computed tomography (CT) and direct radiography (DR). The patient's medical history did not indicate any serious illness, operations, or previous hospitalization. The preoperative diagnosis was multiple craniofacial fractures, facial nerve injury, traumatic cleft palate and brain concussion. The preoperative vital signs were all within normal ranges. Anesthesia was induced with propofol, fentanyl, and rocuronium, and maintained with sevoflurane and remifentanil. Body temperature, blood pressure, respiration and heart rate were normal at the beginning of the operation. By the end of the surgery, the patient's body temperature had increased from 37.1 to 41 °C and his end-tidal carbon dioxide ($ETCO_2$) had risen from 35 to 103 mmHg (Figs. 1 and 2). In the meantime, muscle spasm around the surgical area and limb rigidity could be seen. The procedure was terminated, and the patient was administered pure oxygen whilst undergoing symptomatic treatments, including treatments to correct acidosis, as well as being systemically cooled and undergoing intravenous administration of norepinephrine. Clinical chemistry examinations demonstrated metabolic acidosis, electrolyte disorder and serum enzyme abnormality (Table 1). The electrocardiogram demonstrated a roughly normal pattern 20 min after the onset of MH. Despite all efforts to resuscitate him the patient died 3 h later.

Autopsy findings

At autopsy there were skin bruises around the bilateral orbital, left eyelid conjunctiva congestion, a pale right conjunctiva, and a bilateral nasal blood spill. The central

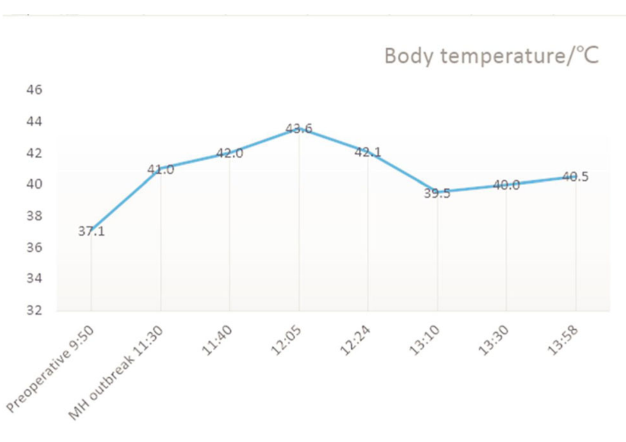

Fig. 1 Continuous observation on the patient's temperature

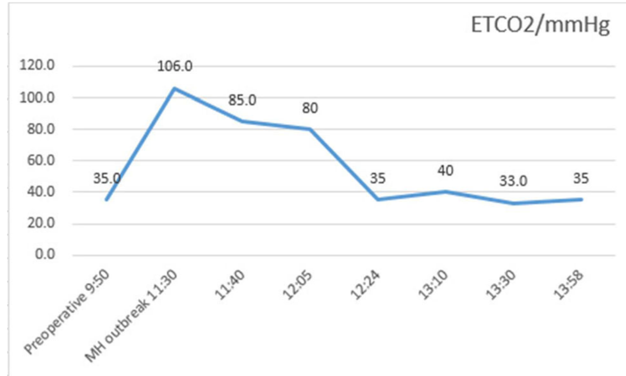

Fig. 2 Monitored ETCO2 of the patient

maxillary incisors and right central incisor were missing and the maxilla and teeth had been fixed with a dental arch splint. Two fresh surgical incisions between the right upper labial frenum and lower labial frenum gingival mucosa had been sutured. Old striped scars and bruises were present on the left cheek and shoulder. There was a hemorrhage on the scalp measuring 7.5 cm × 5.0 cm and a subarachnoid hemorrhage (SAH) measuring 10.0 cm × 8.5 cm, both of which were probably due to the traffic accident. Moreover, a comminuted fracture was present in the sphenoid bone in the anterior cranial fossa. Multiple patchy hemorrhages were seen on the sternohyoid muscle and sternothyroid muscle surface. Apart from the above skeletal muscles, scattered hemorrhages of the heart and lungs accompanied by cellulose exudation were observed. The renal surface and section were basically normal, except for some congestion of the renal medulla. The remainder of the autopsy was unremarkable with no evidence of any significant cardiovascular, gastrointestinal, respiratory, or central nervous system diseases.

Table 1 Results of clinical chemistry examinations

Test items	Results	Unit	Reference values
PH value	6.931		7.35–7.45
PCO_2	129.2	mmHg	35–45
PO_2	196	mmHg	80–100
HCO_3	27.2	mmo1/L	21–27
BE	−8.9	mmHg	−2.5–3
K	9.7	mmo1/L	3.5–5.5
Na	156.4	mmo1/L	135–148
AST	1005.0	U/L	8–40
LDH	2023.0	U/L	109–245
CK	25.0	U/L	25–200
CKMB	784.0	U/L	0–24
a-HBDH	1110.0	U/L	72–182

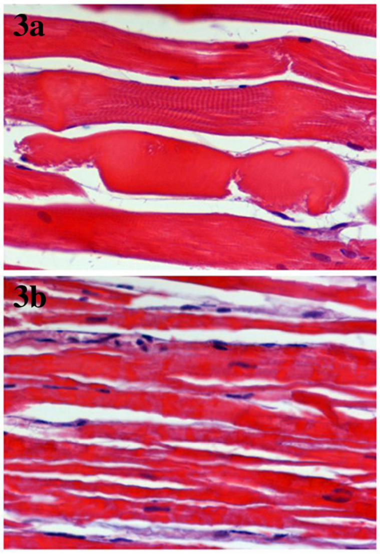

Fig. 3 Histological examination shows eosinophilic changes and focal necrosis of the sternohyoid and sternothyroid skeletal muscles (**a**: HE stain, ×400); Vague striation of myofibril and contraction band necrosis of the left ventricular and papillary muscles (**b**: HE stain, ×400)

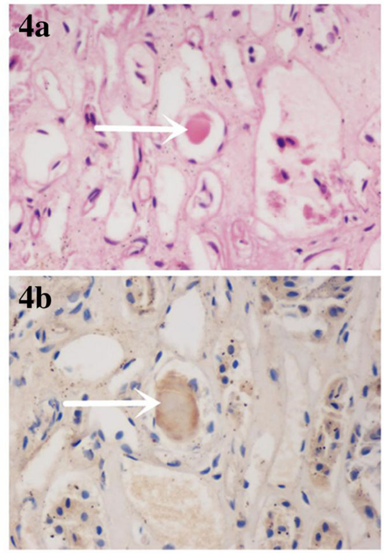

Fig. 4 Shock kidneys with eosinophilic casts in the tubules determined by hematoxylin-eosin staining and immunohistochemical staining (**a**: HE stain, ×400; **b**: IHC stain, ×400)

Histological examination revealed eosinophilic changes and focal necrotic skeletal muscle cells of the sternohyoid and sternothyroid skeletal muscle cells (Fig. 3a), vague striation of myofibrils and contraction bands necrosis of the left ventricular and papillary muscles (Fig. 3b), pulmonary congestion and edema, small alveolar hemorrhages, microthrombus formation in capillaries of the brain and testes, coagulative necrosis of renal tubular epithelial cells, and shock kidneys with eosinophilic casts in the tubules. The cast material stained positive for myoglobin (MB) (Fig. 4). Most of the Purkinje cells were lost or showed pyknotic nuclei and a shrunken eosinophilic homogenized cytoplasm with proliferation of reactive astrocytes in the cerebella.

Transmission electron microscopy (TEM) revealed ultrastructural changes of the masseter muscle, psoas muscle and myocardium. The sarcomere was shortened when the skeletal muscles contracted forcibly and the contraction bands were dramatically visualized by TEM (Fig. 5). The areas of contractions were confined to single myofibers. Sarcomeres in adjacent myofibers, separated from the areas of contraction by the intercalated discs, were markedly overstretched. I bands and H zones were prominent in the overstretched myofibers. In the areas of obvious contractions, focal disruptions and swelling of the sarcolemma were found, especially in areas of extensive myocytolysis and overstretched myofibers. Apart from myofiber contraction, different-shaped necrosis of muscle cells could be seen (Fig. 6).

Genetic analysis

A DNA sequencing library was prepared using the web-based tool Repeat Masker (http://repeatmasker.org). In brief, genomic DNA was extracted from EDTA-anticoagulated peripheral blood using a standard procedure that includes ultrasonication. The chip was prepared with target DNA and subjected to a high-throughput sequencing platform to screen for mutations according to the ClinVar database. The screening genes include ACTA1, ANO5, BAG3, CAPN3, DMD, RYR1, and CACNA1S.

Postmortem genetic analysis revealed that the deceased had a mutation of RYR1, which was found to be heterozygous from an amino acid exchange in exon 11, p.G341A (c.1021G>A). Genetic analysis of family members revealed the same mutation in his father's DNA (Fig. 7).

Discussion

MH is a rare but potentially lethal complication of general anesthesia, with a high mortality rate in the young and in muscular individuals [6]. It manifests a male predominance with a 2:1 male to female ratio [7, 8]. It is a pharmacogenetic disorder in which skeletal muscle membrane abnormalities

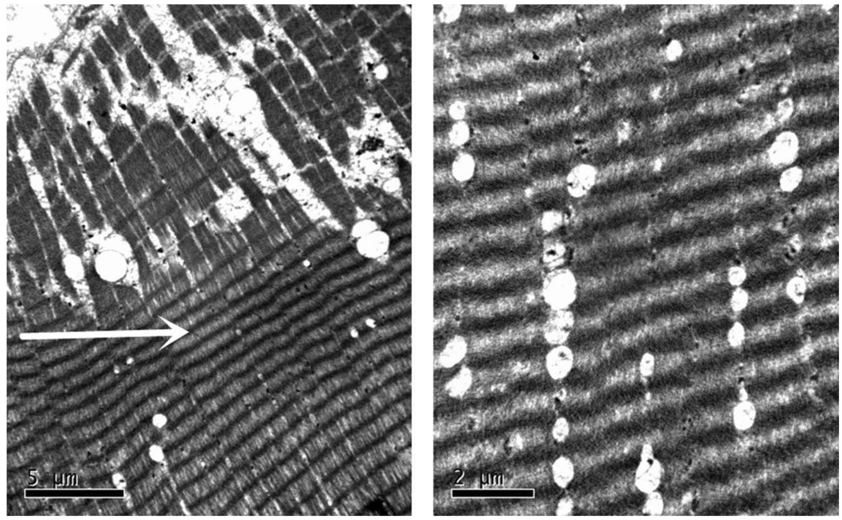

Fig. 5 Sarcomere shorten and contraction bands were dramatically visualized by TEM (white arrow indicates that skeletal muscle forcibly contracted)

adversely influence calcium homeostasis after exposure to a triggering agent such as volatile anesthetics or the depolarizing muscle relaxant succinylcholine [9, 10]. Subsequently, typical symptoms include accelerated skeletal muscle metabolism, muscle rigidity and an abnormally high body temperature [11]. The mortality rate resulting from ventricular fibrillation, pulmonary edema, intravascular coagulopathy, cerebral hypoxic damage, cerebral edema, or renal failure, ranges from 70 to 80% if no immediate treatment is administered [12, 13]. The most important treatment is the discontinuation of anesthesia, and the use of dantrolene, which is regarded as a muscle relaxant in the long term treatment of skeletal muscle spasticity [14]. In this case, the patient died as dantrolene was not available because it is not widely used and routinely stored in Chinese hospitals due to its high price and short shelf life.

The most significant morphological findings in this case were in the skeletal muscles [15, 16]. MH attacks are characterized by muscle spasms and contractures, such as severe masseter muscle and generalized muscular rigidities [17]. Grossly, they may appear pale and swollen, and the corticomedullary junction may be indistinct [16]. It is attributed to heat-related deaths characterized by petechial hemorrhages over the surface of the lungs and heart, visceral congestion, and bleeding from other tissues [18]. The histopathologic changes include hypercontraction bands, segmental necrosis, degenerating fibers and rhabdomyolysis; contraction bands and rhabdomyolysis are especially prominent ultrastructural features [19, 20]. Among the above changes, dissolution of myofilaments and disruption of sarcomeres may be considered to come from prolonged autolysis. However, hypercontraction bands of the skeletal muscles and myocardium are not consistent with changes secondary to autolysis or ischemia [20]. Thereby, hypercontraction bands could be regarded as a specific manifestation of MH. Urinary myoglobin (MB) levels will increase rapidly, associated with hypoxic muscle damage or catabolic muscle changes, and the kidneys normally show eosinophilic casts in the tubules. The cast material in this case

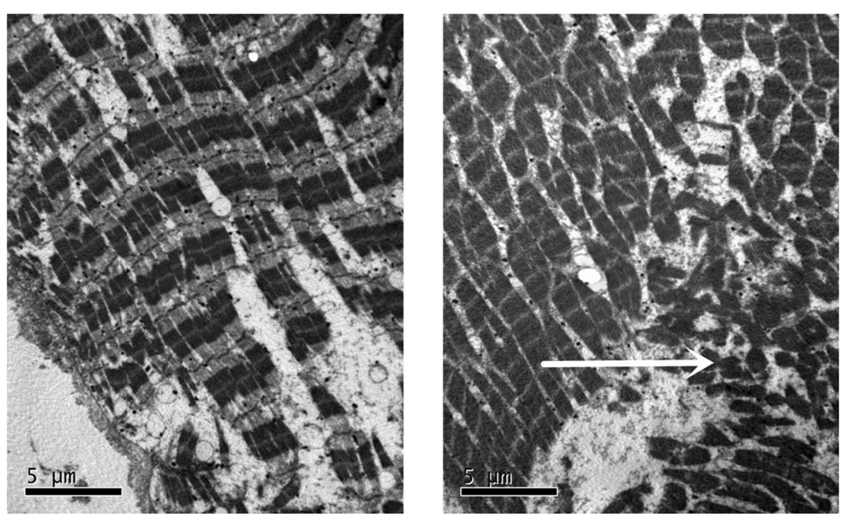

Fig. 6 Different shaped necrosis of muscle cells

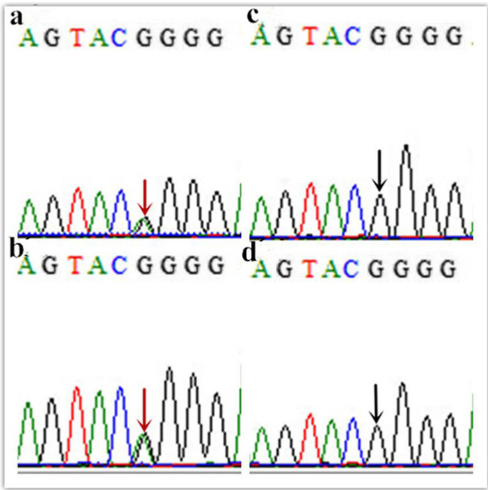

Fig. 7 Genetic analysis was performed and the same mutation was determined in the proband and in his father's DNA (**a**: the proband; **b**: his father; **c**: his mother; **d**: his brother; red arrows reveal the mutation, black arrows point to the normal position)

showed faint, focal positive staining for MB in the tubules as determined by hematoxylin-eosin (HE) staining and immunohistochemistry (IHC) [4]. In this case, the autopsy results and morphology are consistent with the literature. TEM is not indispensable for diagnosing MH, however it provides ultrastructural pathology. In this study contraction band necrosis and rhabdomyolysis were prominent using TEM.

Genetic analyses have revealed that about 50% of families with MH have mutations in the locus of the RYR1 and its encoding gene on chromosome 19q13.1 [3, 21]. The RYR1 contains 106 exons encoding proteins of about 5035 amino acids [22]. Causal mutations of RYR1 for MH include two main regions ranging from amino acid residues 35 to 614 (exons 2 to 17) and 2163 to 2458 (exons 39 to 46) [23]. RYR1 mutations associated with MH consist of G7304A, G7300A, C1840A, C1209G, G1021A, R614C, G2434R and G341R. Among these, the most frequent mutations are R614C, G2434R and G341R [13]. G1021A is a relatively rare variation that was first reported in a Danish family in 1994 [24]. It is estimated that the G1021A causes MH susceptibility (MHS) in 10% of Caucasian MHS cases [25]. G1021A is believed to be MH associated [26–28]. Genetic analysis has been recommended since the identification of the first RYR1 gene associated with MHS in 1990 [29]. Genetic analysis is an attractive proposition to identify variants of skeletal muscle RYR1because it could replace the invasive and morbid in vitro contracture test (IVCT) of a muscle biopsy [28]. Nevertheless, the mutation of RYR1 was systematically investigated to ascertain susceptibility to MH when no muscle contracture test in any relative was conducted [27]. Family mutation screening has potential impacts for the current health of the individual, but it also influences the future health of the proband and their immediate relatives [28]. It is also important to consider that the availability of various forms of family screening will place an additional burden on both the families and their genetic counselors [30]. The cost of performing a muscle biopsy and IVCT is approximately €2000 in Europe. In the USA, the test is the CHCT version and incurs a cost of about $6000 [31]. In the present case, the mutation of RYR1 was confirmed in the deceased's genomic DNA. Subsequently, familial genetic analysis was performed and the same mutation (G1021A) was found in his father's DNA. Therefore, if general anesthesia is required for his father for a future medical treatment, triggering agents such as volatile anesthetics or succinylcholine must be avoided.

The antemortem diagnosis is based on clinical symptoms and laboratory tests [32]. The diagnosis of MH is highly suspected when there is a rise in $ETCO_2$, an elevated body temperature, and an onset of muscle contractures within minutes to hours of administering volatile anesthetics and a depolarizing muscle relaxant. Muscle rigidity, hyperthermia, and metabolic acidosis are the "classic clinical triad" [33]. The postmortem diagnosis of a suspected MH case is normally made difficult by the inconclusiveness of toxicology analysis, autopsy, microscopic findings and the clinical course [15]. The differential diagnosis of MH includes malignant neuroleptic syndrome, neuromuscular disorders, elevated ETCO2 due to laparoscopic surgery, infection or septicemia [34]. Currently, genetic analysis is performed using a Sanger sequencing with QIAamp DNA Blood Midi Kit (Qiagen, Hilden, Germany) to identify the candidate mutations. Sanger sequencing costs about 2000 RBM (about 300 US dollars) per sample. Therefore, genetic analysis is a cost-effective and helpful way to confirm MH.

In conclusion, this is the first report in forensic medicine of a RYR1 mutation in both a healthy young man who died of MH, and in a member of his family. The forensic postmortem diagnosis is based on clinical symptoms, autopsy, a thorough microscopic examination, and genetic analysis. Therefore, postmortem genetic analysis, i.e. molecular autopsy, becomes an important tool to confirm the mutation of RYR1 in MH and to identify pre-symptomatic patients and asymptomatic carriers. It provides an opportunity to assess potential risks for family members of the deceased and to diminish the occurrence of malignant events.

Key points

1. We report a case of malignant hyperthermia with a mutation of RYR1.
2. Malignant hyperthermia was diagnosed via an autopsy, morphology, and genetic analysis.
3. Genetic analysis and mutations of RYR1 are discussed.

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Compliance with ethical standards

Conflict of interest None.

Informed consent This study was approved by the Ethics Committee of Tongji Medical College, Huazhong University of Science and Technology, Wuhan, Hubei, China, and written informed consent was obtained from the family members of the patient.

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